

Customer Load Violation Model

Current Findings

21 September 2026

1 Mathematical formulation

N numbers of customers are divided into small (S), medium (M), and large (L) classes. Let n_S, n_M, n_L denote the number of customers in each class and let I_S, I_M, I_L denote their respective current demands. This means:

$$n_S + n_M + n_L = N, \quad n_S, n_M, n_L \in \{0, 1, \dots, N\}. \quad (1)$$

And for customer i in class $j \in \{S, M, L\}$, define the state $X_{j,i} \in \{0, 1\}$, where 1 means on and 0 means off. The aggregate current is:

$$A = \sum_{j \in \{S, M, L\}} I_j \sum_{i=1}^{n_j} X_{j,i}. \quad (2)$$

A violation occurs when the aggregate current is strictly above the network limit T . Its probability is:

$$\mathbb{P}(A > T). \quad (3)$$

The current study uses:

$$(I_S, I_M, I_L) = (20, 40, 100), \quad N = 40, \quad T = 1000. \quad (4)$$

2 All-On model

Assuming all customer loads are on, every state variable is fixed at $X_{j,i} = 1$, so the aggregate current for a given composition is:

$$A_{\text{all}} = I_S n_S + I_M n_M + I_L n_L. \quad (5)$$

Consequently, the violation probability for a fixed composition is

$$\mathbb{P}(A > T \mid n_S, n_M, n_L, \text{all on}) = \begin{cases} 1, & I_S n_S + I_M n_M + I_L n_L > T, \\ 0, & I_S n_S + I_M n_M + I_L n_L \leq T. \end{cases} \quad (6)$$

The graph demonstrates the violating and non-violating composition points for the parameters $I_S = 20$, $I_M = 40$, $I_L = 100$, $N = 40$, and $T = 1000$.

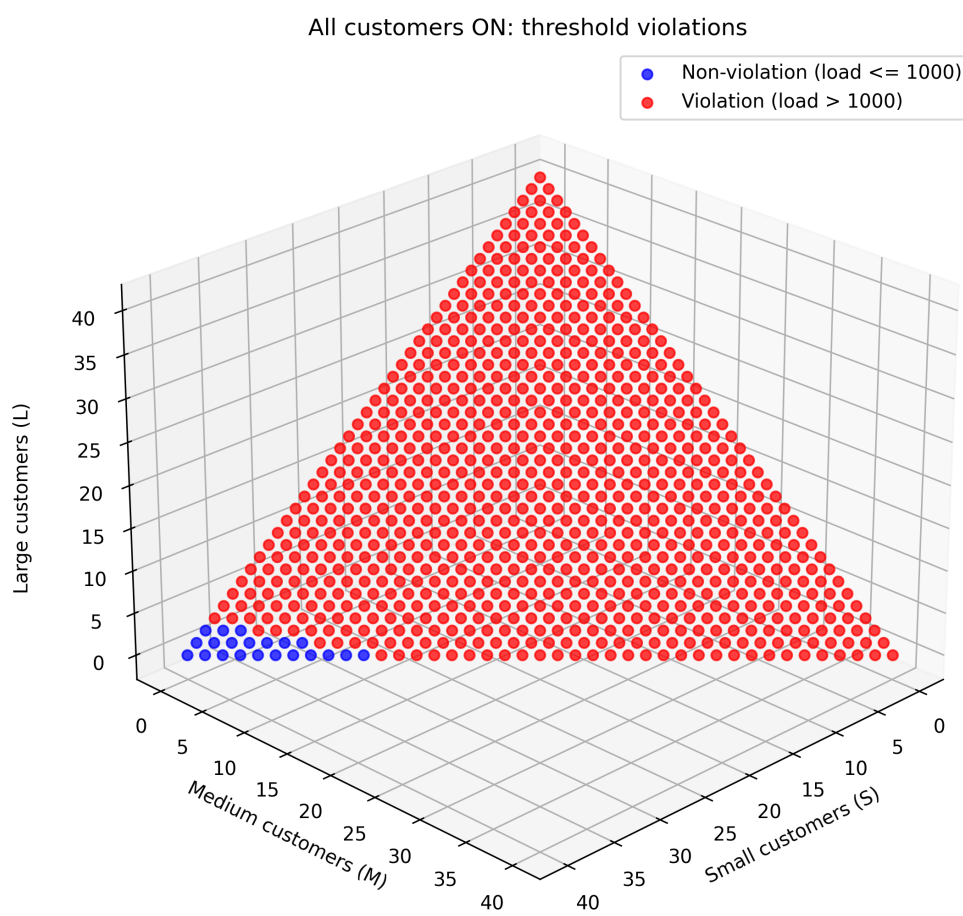


Figure 1: All-On model results.

To count the total number of compositions, each composition corresponds to a different placement of the two separators in the array:

$$\underbrace{\star \cdots \star}_{n_S} \mid \underbrace{\star \cdots \star}_{n_M} \mid \underbrace{\star \cdots \star}_{n_L} \quad .$$

Choosing the two separator positions among $N + 2$ positions gives

$$\text{Number of compositions} = \binom{N+2}{2} = \frac{(N+2)(N+1)}{2}. \quad (7)$$

Empty groups are allowed. For $N = 40$, this gives $\binom{42}{2} = 861$ compositions. The model finds 840 violating compositions.

$$\mathbb{P}(A > T \mid \text{uniform composition, all on}) = \frac{840}{861} = 0.9756 \quad (97.56\%). \quad (8)$$

3 On–Off model

The On–Off model treats customers as independent Bernoulli loads. In the current run,

$$X_{j,i} \stackrel{\text{ind}}{\sim} \text{Bernoulli}(p), \quad p = 0.5. \quad (9)$$

For each class j , define Y_j as the number of active customers in that class. It is the sum of the individual on–off states and therefore has a binomial distribution:

$$Y_j = \sum_{i=1}^{n_j} X_{j,i} \sim \text{Binomial}(n_j, p), \quad j \in \{S, M, L\}. \quad (10)$$

The random aggregate current in one trial is then

$$A = I_S Y_S + I_M Y_M + I_L Y_L. \quad (11)$$

The implemented model estimates the probability of violations by Monte Carlo simulation. For each of the 861 feasible compositions, it performs the following calculation independently:

1. draw Y_S, Y_M, Y_L from their binomial distributions;
2. calculate $A = I_S Y_S + I_M Y_M + I_L Y_L$;
3. record whether $A > T$; and
4. run 100,000 trials for each of five seeds (42–46), giving $N_{\text{trials}} = 500,000$ pooled trials per composition.

If $N_{\text{violations}}$ trials violate the limit, the reported estimator is

$$\hat{\mathbb{P}}(A > T) = \frac{N_{\text{violations}}}{N_{\text{trials}}}. \quad (12)$$

Across all 861 equally weighted compositions, the overall Monte Carlo violation probability is $213,644,734/430,500,000 \approx 49.62712\%$.

Monte Carlo: $P(\text{load} > 1000)$
 $p(\text{on}) = 0.5$; 5 seeds x 100,000 trials = 500,000 per combination

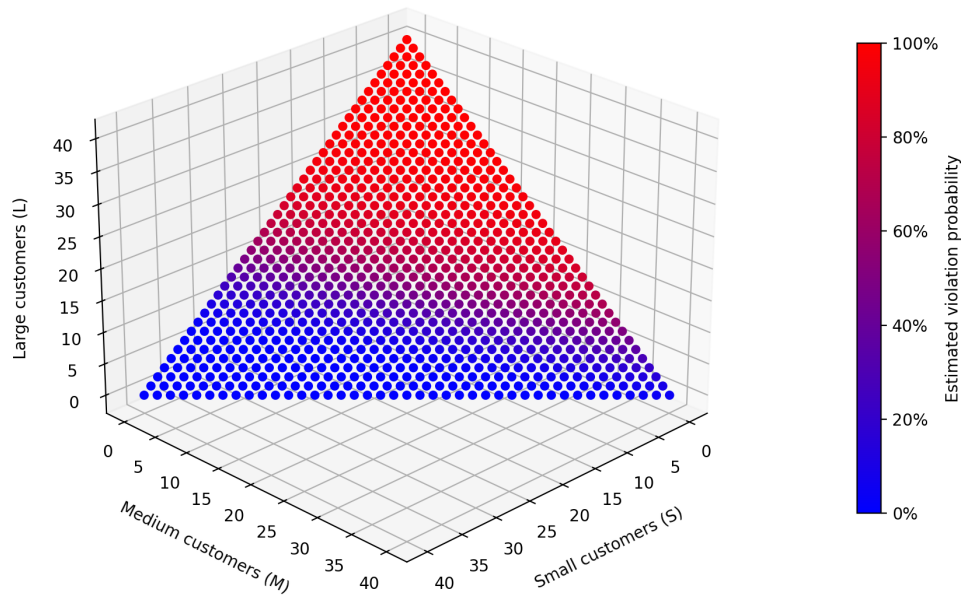


Figure 2: Estimated On-Off violation probability for every customer composition, pooled across five seeds with 100,000 trials per seed.